

DSS5202 Sustainable Systems Analysis

Assignment #2

Due: 14 September 2026, 9 pm

Instructions

You must show your workings before using computing tools to compute the final answers.

Show the formula that is being computed or the equation to be solved.

A direct copy-and-paste from Excel or Python financial function is not acceptable.

All diagrams where required may be hand-drawn.

Submit your answers as one PDF file via Canvas.

Problem Description

A university plans to develop an electric-vehicle (EV) charging facility on campus to provide paid charging services to staff, students, and visitors. The project will support low-carbon transportation and promote the adoption of electric vehicles. The university is considering the following two mutually exclusive alternative systems that provide the same charging capacity:

- A. Grid-powered EV charging system** where all the electricity required by the charging equipment will be purchased from the public electricity grid.
- B. Hybrid Solar-Grid EV charging system** where a solar photovoltaic panels canopy will be installed above the parking area. Solar generated electricity will be used primarily for vehicle charging and any additional electricity required will be purchased from the public grid.

Both alternatives will have a useful life of 15 years. Relevant information for both systems is provided in the table below.

Revenue or Cost	System A Grid-powered	System B Hybrid Solar-Grid
Initial acquisition and installation cost at time 0	\$1,200,000	\$2,000,000
Annual charging-service revenue, Years 1–8	\$520,000	\$520,000
Annual charging-service revenue, Years 9–15	\$470,000	\$470,000
Annual grid-electricity cost, Years 1–8	\$220,000	\$80,000
Annual grid-electricity cost, Years 9–15	\$245,000	\$105,000
Annual operation and maintenance cost, Years 1–15	\$55,000	\$75,000
Salvage value at the end of Year 15	\$60,000	\$110,000

The university's infrastructure development office has indicated that the minimum attractive rate of return (*MARR*) for this project is 10%, and a study period of 15 years will be used in the analysis.

Question 1.

Draw a cash-flow diagram for each alternatives over the 15-year study period. (10 marks)

Question 2.

Use the Net Present Value (*NPV*) method to determine which alternative should be selected.

(10 marks)

Question 3.

Use the Internal Rate of Return (*IRR*) method to determine which alternative should be selected.

(10 marks)

Question 4.

Determine the range of *MARR* for which the following is statement true:

$$NPV(MARR) \text{ of A} > NPV(MARR) \text{ of B} \quad \text{if and only if} \quad IRR \text{ of A} > IRR \text{ of B}$$

where A and B stands for System A and System B, respectively.

Explain your answer.

(10 marks)

Question 5.

Using the Modified Internal Rate of Return (*MIRR*) method determine if System B is financially feasible if the financing rate is 8% and the reinvestment rate is 10%. Explain your answer.

(10 marks)

***** End of Assignment *****